Prayas JEE AIR (2025)

WEEKLY TEST 24

DURATION : 180 Minutes | **DATE : 26/12/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS		CHEMISTRY		MATHEMATICS	
1.	(2)	26.	(2)	51	(2)
2.	(1)	27.	(4)	52.	(1)
3.	(1)	28.	(2)	53.	(3)
4.	(4)	29.	(2)	54.	(4)
5.	(4)	30.	(4)	55.	(2)
6.	(3)	31.	(3)	56.	(4)
7.	(4)	32.	(4)	57.	(4)
8.	(3)	33.	(2)	58.	(1)
9.	(1)	34.	(4)	59.	(4)
10.	(1)	35.	(3)	60.	(1)
11.	(2)	36.	(4)	61.	(1)
12.	(1)	37.	(1)	62.	(3)
13.	(2)	38.	(2)	63.	(4)
14.	(3)	39.	(2)	64.	(2)
15.	(2)	40.	(3)	65.	(1)
16.	(2)	41.	(4)	66.	(4)
17.	(4)	42.	(1)	67.	(1)
18.	(3)	43.	(1)	68.	(2)
19.	(3)	44.	(1)	69.	(2)
20.	(4)	45.	(3)	70.	(3)
21.	(100)	46.	(3)	71.	(63)
22.	(180)	47.	(0)	72.	(10)
23.	(625)	48.	(3)	73.	(40)
24.	(10)	49.	(2)	74.	(10)
25.	(40)	50.	(1)	75.	(1)

SECTION-I (PHYSICS)

1. (2)

At maximum height, velocity is zero so,

$0 = u^2 - 2gh \;\; or \;\; u = \sqrt{2gh}$

As $s = ut + \frac{1}{2}at^2$ so, $\frac{h}{3} = \sqrt{2ght} - \frac{1}{2}gt^2$

$gt^2 - 2\sqrt{2ght} + \frac{2h}{3} = 0$

or $t = \frac{2\sqrt{2gh} \pm \sqrt{8gh - \frac{8gh}{3}}}{2g}$

$\frac{t_1}{t_2} = \frac{2\sqrt{2gh} - \sqrt{8gh - \frac{8gh}{3}}}{2\sqrt{2gh} + \sqrt{8gh - \frac{8gh}{3}}} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

2. (1)

Let additional force is $\vec{F}$, so that $F_{net} = 0$

$\vec{F}' + 5(+\hat{i}) + 6(-\hat{i}) + 7(+\hat{j}) + 8(-\hat{j}) = 0$

$\Rightarrow \;\; \vec{F}' - (\hat{i} + \hat{j}) = 0 \Rightarrow \vec{F}' = \hat{i} + \hat{j}$

$\Rightarrow \;\; |\vec{F}| = \sqrt{1^2 + 1^2} = \sqrt{2}N$

$\theta = \tan^{-1}\left(\frac{\text{Component of } \hat{j}}{\text{Component of } \hat{i}}\right)$

$\theta = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ$

3. (1)

We know that $P = mV = \sqrt{2mKE}$

$\therefore \;\; 0.2u = \sqrt{2 \times 0.2 \times 90}$

$\Rightarrow \;\; u = 30\,\text{m/s}$

and $0.2v = \sqrt{2 \times 0.2 \times 40}$

$\Rightarrow \;\; v = 20\,\text{m/s}$

So, $a = \frac{v - u}{t} = \frac{20 - 30}{1} = \frac{-10\,\text{ms}^{-2}}{s^2}$

Now, using $x^2 - u^2 = 2aS$

$\Rightarrow \;\; S = \frac{0 - 30^2}{2 \times (-10)} = 45\,\text{m}$

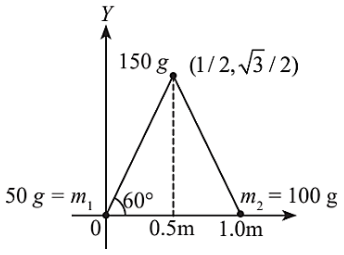
4. (4)

$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{m_1 + m_2 + m_3}$

$\vec{r}_{cm} = \frac{0 + 150 \times \left(\frac{1}{2}i + \frac{\sqrt{3}}{2}\hat{j}\right) + 100 \times \hat{i}}{300}$

$\vec{r}_{cm} = \frac{7}{12}\hat{i} + \frac{\sqrt{3}}{4}\hat{j}$

$\therefore$ Co-ordinate of center $= \left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)m$



5. (4)

$I_X = I_Y = \frac{MR^2}{4}$

If radius of gyration is K_1

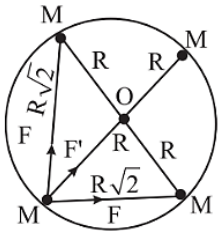
then $I_x = I_y = MK_1^2 \Rightarrow K_1 = \frac{R}{2}$

similarly, $I_Z = \frac{MR^2}{2} = MR_{K_2}^2 \Rightarrow K_2 = \frac{R}{\sqrt{2}}$

Hence, $K_1 \neq K_2$.

6. (3)

$F_{net} = \frac{Mv^2}{R}$



$\sqrt{2}F + F' = \frac{Mv^2}{R}$

$\sqrt{2}\frac{GMM}{(\sqrt{2}R)^2} + \frac{GMM}{(2R)^2} = \frac{Mv^2}{R}$

$\frac{GM}{R}\left[\frac{1}{\sqrt{2}} + \frac{1}{4}\right] = v^2$

$v = \sqrt{\frac{GM(4 + \sqrt{2})}{R4\sqrt{2}}}$

$v = \frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2} + 1)}$

7. (4)

$Y = \frac{F/A}{\frac{\Delta l}{l}}$

$\Rightarrow \;\; F = \frac{YA}{l}\Delta l$

$\left(\frac{A\Delta l}{l}\right)_1 = \left(\frac{A\Delta l}{l}\right)_2$

$$\Rightarrow \frac{\Delta l_2}{\Delta l_1} = \frac{A_1}{A_2} \times \frac{l_2}{l_1}$$

$$\Rightarrow \frac{\Delta l_2}{0.2} = \frac{1}{2.4 \times 2.4} \times \frac{2}{1}$$

$$\Rightarrow \Delta l = 6.9 \times 10^{-2} \text{ mm}$$

8. (3)

A. Torque $[\tau] = [F][r_\perp]$

$$[\tau] = \left[MLT^{-2}\right][L] \Rightarrow \left[ML^2T^{-2}\right]$$

B. $[\text{Stress}] = \frac{[F]}{[A]} = \frac{MLT^{-2}}{L^2} = \left[ML^{-1}T^{-2}\right]$

C. $[\text{Pressure gradient}] = \frac{[\Delta P]}{[\Delta X]} = \frac{\left[ML^{-1}T^{-2}\right]}{L}$

$$\Rightarrow \left[ML^{-2}T^{-2}\right]$$

D. Coefficient of viscosity $\Rightarrow F = 6\pi\eta rv$

$$[\eta] = \frac{[F]}{[r][v]} = \frac{\left[MLT^{-2}\right]}{[L]\left[LT^{-1}\right]}$$

$$[\eta] = ML^{-1}T^{-1}$$

9. (1)

Using equation, PV^γ g= Constant,

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^\gamma$$

$$\frac{81}{16} = \left(\frac{27}{8}\right)^\gamma$$

$$\left(\frac{3}{2}\right)^4 = \left(\frac{3}{2}\right)^{3\gamma}$$

$$\Rightarrow \gamma = \frac{C_P}{C_V} = \frac{4}{3}$$

10. (1)

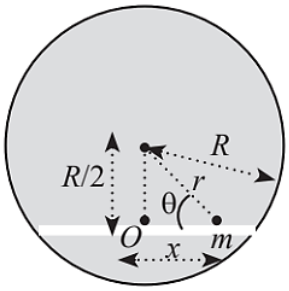
Force, F = m Ecosθ

$$= m\left(\frac{GM}{R^3} \times r\right) \times \frac{x}{r}$$

$$= \frac{mgR^2}{R^3} \times x$$

$$ma = \frac{mg}{R}x \Rightarrow a = \frac{g}{R}x$$

Time period of SHM, $T = 2\pi\sqrt{\frac{R}{g}}$



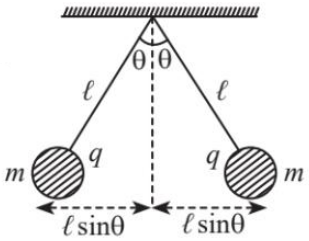
O ⇒ Mean Position

11. (2)

Distance between the charges $r = \ell\sin\theta + \ell\sin\theta$

Applying Newton's second law,

$$T\sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2\ell\sin\theta)^2}$$

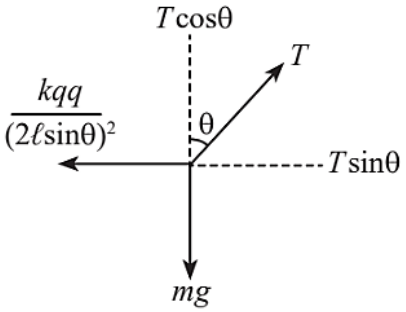


$$T\cos\theta = mg \therefore \tan\theta = \frac{q^2}{4\pi\varepsilon_0 mg\, 4\ell^2\theta^2}$$

[For small angle, $\tan\theta \approx \theta$] *So,* $\theta^3 = \frac{q^2}{16\pi\varepsilon_0 mg\ell^2}$

$$\theta = \left(\frac{q^2}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3}$$

Also separation $x = 2\ell\sin\theta \cong 2\ell\theta$



$$x = 2\ell\left(\frac{q^2}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3}$$

$$x = \left(\frac{8q^2\ell^3}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3}$$

$$x = \left(\frac{q^2\ell}{2\pi\varepsilon_0 mg}\right)^{1/3}$$

12. **(1)**

$\Sigma q = 0$

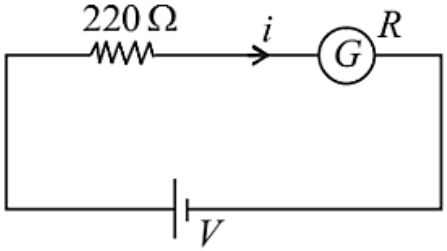
$\Rightarrow \quad V = 0$

Two regular pentagons of positive and negative charge will from electric field due to each pentagon is zero.

13. **(2)**

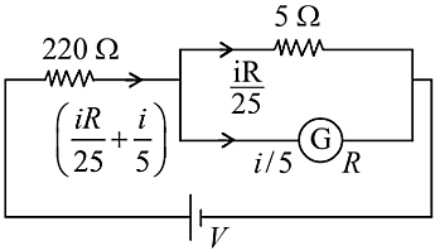
Case 1: K_1 and K_2 are open.

$$i = \frac{V}{220 + R}$$



Case 2: K_1 and K_2 are closed.

$$\left(\frac{iR}{25} + \frac{i}{5}\right)220 + \frac{i}{5}R = V$$



$$\left(\frac{iR}{25} + \frac{i}{5}\right)220 + \frac{i}{5}R = i(220 + R)$$

After solving $R = 22\Omega$

14. **(3)**

$$B_C = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2\left(x^2 + r^2\right)^{3/2}}$$

At $x = \frac{r}{2}$

$$B_a = \frac{\mu_0 I r^2}{2\left(\frac{r^2}{4} + r^2\right)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2\left(\frac{5}{4}r^2\right)^{3/2}} = \frac{\mu_0 I}{2r}\left(\frac{4}{5}\right)^{3/2}$$

$$= \frac{\mu_0 I}{2r}\left(\frac{2}{\sqrt{5}}\right)^3 = \left(\frac{2}{\sqrt{5}}\right)^3 B$$

15. **(2)**

$I = I_0 t - I_0 t^2$

$\phi = BA$

$\phi = \mu_0 nIA$

$$V_R = \frac{d\phi}{dt} = -\mu_0 nAI_0(1 - 2t)$$

$V_R = 0$ at $t = \frac{1}{2}$ and $I_R = \frac{V_R}{\text{Resistance of loop}}$

16. **(2)**

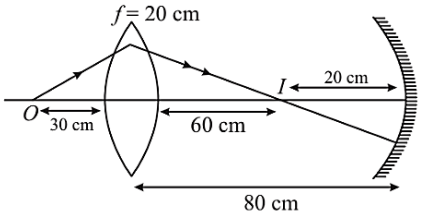


Image formed by lens

$$\frac{1}{v} + \frac{1}{30} = \frac{1}{20} \qquad \Rightarrow v = +60\,\text{cm}$$

If image position does not change even when mirror is removed it means image formed is at centre of curvature of the mirror.

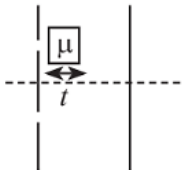
Radius of curvature of mirror = 80 – 60 = 20 cm

$\Rightarrow \quad f = 10$ cm

For virtual image, object is to be kept between focus and pole.

$\Rightarrow$ Maximum distance of object from spherical mirror for which virtual image is formed, is 10 cm.

17. **(4)**



$\Delta X = (\mu - 1)t = 1\lambda$

For one maximum shift $t = \frac{\lambda}{\mu - 1}$

18. **(3)**

$\vec{E} = E_0\hat{n}\sin(\omega t + (6y - 8z)) = E_0\hat{n}\sin(\omega t + \vec{k}\cdot\vec{r})$

where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{k}\cdot\vec{r} = 6y - 8z$

$\Rightarrow \vec{k} = 6\hat{j} - 8\hat{k}$

Direction of propagation $\hat{s} = -\hat{k} = \left(\frac{-3\hat{j} + 4\hat{k}}{5}\right)$

19. **(3)**

$M_0 = 200\text{MeV}/\text{C}^2, m = 1\text{Gev}/\text{C}^2$

Initial velocity of particle is 'V_0'.

Final velocity of hydrogen atom is 'V'

$M_0V_0 = mV$

$V = \dfrac{M_0V_0}{m}$

Also, $\dfrac{1}{2}M_0V_0^2 = \dfrac{1}{2}mV^2 + \dfrac{3}{4}\times 13.6$

Put (i) and get answer

$\dfrac{1}{2}M_0V_0^2 = \dfrac{51}{4}\text{eV}$

Hence, $N = 51$.

20. **(4)**

As a neutron has more rest mass than a proton, so proton will require energy to decay into neutron.

21. (**100**)

Velocity of wave on string $= \sqrt{\dfrac{T}{\mu}}$

= 8 m/s

The pulse gets inverted after reflection from the fixed end, so for constructive interference to take place between successive pulses, the first pulse has to undergo two reflections from the fixed end.

$\Delta t = \dfrac{2L}{v} = \dfrac{2\times 0.4}{8} = 0.1s = 100$ ms.

22. **(180)**

Given, $\vec{E} = (Ax + B)\hat{i}$ *or* $E = 20x + 10$

Using $V = \int E dx$

we have $V_2 - V_1 = \int_{-5}^{1}(20x + 10)dx = -180\,\text{V}$

or $V_1 - V_2 = 180\ V$

23. (**625**)

For the maximum current,

$X_C = X_L$

or $\dfrac{1}{\omega C} = \omega L$

$\therefore \quad C = \dfrac{1}{\omega^2 L} = \dfrac{1}{(2\pi f)^2 L}$

$= \dfrac{1}{\left(2\times 2\times 10^3\right)^2 \times 100\times 10^{-3}}$

$= 625\times 10^{-9}\ F = 625nF$

24. **(10)**

de-Broglie wavelength,

$\lambda = \dfrac{\text{h}}{\text{mv}} = 663\left(\dfrac{3\times 10^8}{27.3\times 10^{14}}\right) \quad \left[\because \lambda = \dfrac{\text{c}}{\text{v}}\right]$

$v = \dfrac{6.63\times 10^{-34}\times 27.3\times 10^{14}}{9.1\times 10^{-31}\times 3\times 10^8\times 663}$

$v = 10\ m/\text{sec}.$

25. **(40)**

$M_{\text{ice}}L_f + M_{\text{ice}}(40-0)C_W = M_{\text{steam}}\,L_V + M_{\text{steam}}$

$(100-40)C_w$

$\Rightarrow \quad 200[80 + 40(1)] = \text{M}(540 + 60(1))$

$\Rightarrow \quad 200(120) = M(600)$

$M = 40\ \text{gm}.$

SECTION-II (CHEMISTRY)

26. **(2)**

Degenerate orbitals must have same energy.

∴ Orbitals with same values of n and l are degenerate

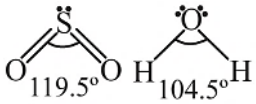
27. **(4)**

$2CCl_3CHO \xrightarrow{OH} CCl_3COONa + CCl_3CH_2OH$

Cannizzaro reaction is a disproportionation reaction. One aldehyde molecule is oxidized to salt of the carboxylic acid, other one is reduced to alcohol. So, the compound is CCl_3CH_2OH. IUPAC name is 2, 2, 2, -trichloro ethanol.

28. **(2)**

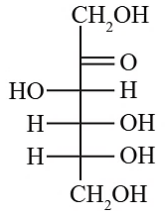
The structures of both SO_2 and H_2O are bent or V-shaped. These are shown as,



Bond angle of $SO_2(sp^2)$ is greater than that of $H_2O(sp^3)$ due to higher repulsion of multiple bonds.

29. (2)

Fructose is a ketohexose.



30. (4)

Rate equation is to be derived wrt slow step

$\therefore$ from mechanism (i)

Rate = $k[Cl_2]\,[H_2S]$

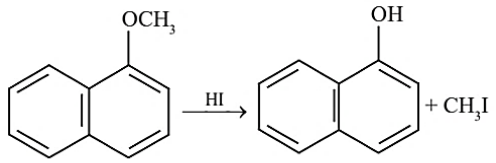
31. (3)

$8 \times 10^{-12} = (2S' + 0.1)^2 S'$

or $S' = 8 \times 10^{-10}$ M

32. (4)

The given reaction occurs as:



1- methoxy naphthalene

33. (2)

Van't Hoff's factor (i) for $Na_2SO_4 = 3$

$\therefore \quad \Delta T_f = (i)k_f\, m$

$= 3 \times 1.86 \times \dfrac{0.01}{1} = 0.0558$ K

34. (4)

Spin only magnetic moment formula is: $\mu = \sqrt{n(n+2)}$ B.M,

n = number of unpaired electrons

$_{22}Ti^{+3} = [Ar]4s^0 3d^1$, unpaired electron =1;

μ =1.73 B.M)

$_{22}V^{+2} = [Ar]4s^0 3d^3$, unpaired electron & =3;

μ =3.87 B.M)

$_{21}Sc^{+3} = [Ar]4s^0 3d^0$, unpaired electron

$= 0; \mu = 0$ B.M

35. (3)

The units of the given parameters can be calculated as:

(1) Cell constant $= \left(\dfrac{\ell}{A}\right) = m / m^2 = m^{-1}$

(2) Molar conductivity $(\lambda_m) = \dfrac{\kappa \times 1000}{\text{molarity}}$

$= Scm^2\, mol^{-1}$

(3) Conductivity $(\kappa) = \dfrac{1}{\rho} = \dfrac{\ell}{RA}$

$= \Omega^{-1} m^{-1}$ or Sm^{-1}

(4) Degree of dissociation

$= \dfrac{\text{number of mole dissociated}}{\text{Total mole}}$

it is dimensionless

36. (4)

Mass of pure C in coal

$= 0.6 \times 1000\,g \times \dfrac{60}{100} = 360\,g$

Mass of C converted into CO

$= \dfrac{60}{100} \times 360 = 216\,g$

$\Rightarrow$ Mol $= \dfrac{216}{12} = 18$ mole of C

Mass of C converted into CO_2 = 360–216 = 144g

$\Rightarrow$ Mole $= \dfrac{144}{12} = 12$ mole of C

$C(s) + O_2(g) \rightarrow CO_2(g) + 400\,kJ$

Mole =12 for CO_2 production, Hence the total energy produced = 400 × 12 = 4800 kJ

$= 400 \times 12 = 4800\,kJ$

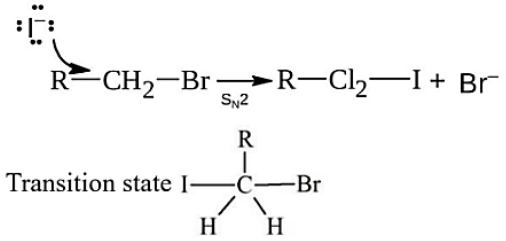
$C(s) + \dfrac{1}{2}O_2(g) \rightarrow CO(g) + 100\,kJ$

Mole =18 for CO production, Hence, the energy produced =100×18=1800 kJ

Thus, total heat =4800 + 1800 = 6600 kJ

37. (1)

This is Finkelstein reaction



Clearly, the transition state is less polar than free anions. Br^- and I^-

Acetic acid is protic which does not support S_N2

Acetone does not solvate anion Br^- gets precipitated and hence can not compete with I^-

So only (a) is correct

38. (2)

According to question,

M_{eq} of Base $= M_{eq}$ of Acid

For H_3PO_3; $1 \times V = 50 \times 1 \times 2$

$\therefore$ V=100 mL of NaOH

For H_3PO_2; $1 \times V = 100 \times 2 \times 1$

$\therefore$ V = 200 mL of NaOH

39. (2)

A complex having strong field ligand has tendency to absorb light of highest energy. Among the three complexes.

$[Co(NH_3)_6]^{+3}$ will absorb radiation of highest energy and least wavelength.

$[Co(NH_3)_5H_2O]^{+3}$ has field weaker than the above compound and therefore absorb radiation of lesser energy and more wavelength.

$[CoCl(NH_3)_5]^{+2}$ has the weakest field and therefore will absorb light of least energy and highest wavelength.

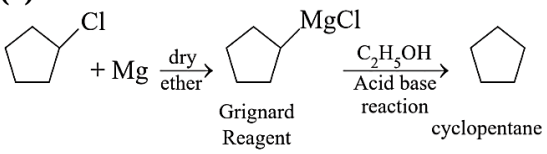
Strength of ligand $NH_3 > H_2O > Cl$.

40. (3)

On moving left to right in a period, atomic number as well as acidic character of oxides increase. 3rd period element is oxides are given as,

$\underbrace{Na_2O \quad MgO}_{\text{Basic}} \quad \underbrace{Al_2O_3}_{\text{Amphoteric}} \quad \underbrace{SiO_2 \quad P_2O_5 \quad Cl_2O_7}_{\text{Acidic}}$

41. (4)



42. (1)

Chlorine is a highly reactive element and it can easily combine with oxygen and can form ClO, ClO_2, $Cl_2O_9Cl_2O_7$ and it has highly oxidising agent and tend to explode.

Reactivity of chlorine can be determined by ability to react with oxygen and halogen.

43. (1)

$$\text{Basicity} \propto \frac{1}{\text{Electronegativity}}$$

Therefore, order would be:

i > iv > ii > iii > v.

44. (1)

Acidity $\propto$ Stability of carbanion (formed by removal of H-atom)

$$\text{Stability of carbanion} \propto \frac{\text{E.W.G.}(-NO_1)}{\text{E.D.G.}(-OMe)}$$

Also, the resonance stabilised carbanion are more stable.

Hence, the correct order is E > C > D > A > B for stability of carbanion and as well as acidity.

45. (3)

Gabriel phthalimide synthesis gives 1° amine in good yield.

46. (3)

$NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(s)$

Let order w.r.t. to [NO] be m and that of $[Cl_2]$ be n.

$rate = k[NO]^m[Cl_2]^n$

$rate_1 = k(0.1)^m(0.1)^n$

$rate_2 = k(0.1)^m(0.2)^n$

$rate_3 = k(0.2)^m(0.2)^n$

we have, $\frac{0.35}{0.18} = \frac{k(0.1)^m(0.2)^n}{k(0.1)^m(0.1)^n}$

On solving

$2 = 2^n$

n =1

we have $\frac{1.40}{0.35} = \frac{k(0.2)^m(0.2)^n}{k(0.1)^m(0.1)^n}$

On solving,

$4 = 2^m$

$2^2 = 2^m$

m = 2

Thus, Overall order $\Rightarrow$ m + n = 2 + 1 = 3

47. (0)

50% of 0.02 mmol and 1 L of H_2SO_4 = 0.01 m mol

Total moles of H_2SO_4 after adding 0.01 mol of H_2SO_4

$=0.01 + 0.01 = 0.02$mmol

$=0.00002 \times 10^3$ mmol

48. (3)

In $[NiCl_4]^{2-}$, n = 2

$\mu = \sqrt{n(n+2)}$BM

$= \sqrt{2(2+2)} = 2.82$ BM

49. **(2)**

Aldehyde is oxidized to carboxylic acid salt in Tollen's test for aldehyde as:

$R–CHO + H_2O \longrightarrow R–COO^- + 3H^+ + 2e^-$

Hence, $2e^{-1}$ s are transferred per aldehyde group.

50. **(1)**

According to Raoult's low of partial pressure,

$\frac{P^0 - P_s}{P_s} = \frac{n_{solute}}{n_{solvent}}$ where,

where, P^0 = vapour pressure of P_s = vapour pressure of solution. pure solvent n = no. of moles

$$= \frac{\frac{x}{60}}{\frac{1000}{18}} = \frac{P^0 - 0.75P^0}{0.75P^0}$$

$$\Rightarrow x = \frac{10000}{9} = 1111.11 \approx 1111 \text{ gm}$$

SECTION-III (MATHEMATICS)

51 **(2)**

$\alpha = \omega, b = 1 + \omega^3 + \omega^6 + \ldots = 101$

$a = (1+\omega)\left(1 + \omega^2 + \omega^4 + \ldots \omega^{198} + \omega^{200}\right)$

$$= (1+\omega)\frac{\left(1 - \left(\omega^2\right)^{101}\right)}{1-\omega^2} = \frac{(1+\omega)(1-\omega)}{1-\omega^2} = 1$$

Equation: Equation: $x^2 - (101 + 1)x + (101)x\ 1 = 0$

$\Rightarrow x^2 - 102x + 101 = 0$

52. **(1)**

Let $E = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, F = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$

$$FE = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} F = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$B = EAF$

$$B^n = \underbrace{(EAF)(EAF)(EAF)\ldots\ldots(EAF)}_{n\text{-iime}}$$

$$\Rightarrow B^n = EA^nF \left[\because CADCAD = CA^2D\right] \ldots..(i)$$

$$A^2 = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{2}{51} \\ 0 & 1 \end{bmatrix} \quad A^3 = \begin{bmatrix} 1 & \frac{3}{51} \\ 0 & 1 \end{bmatrix}$$

Similarly, $A^n = \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}$

$$B^n = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{n}{51}+2 \\ -1 & -\frac{n}{51}-1 \end{bmatrix}\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{n}{51}+1 & \frac{n}{51} \\ -\frac{n}{51} & 1-\frac{n}{51} \end{bmatrix}$$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 25+50 & 25 \\ -25 & -25+50 \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

$\left[\because \Sigma n = \frac{n(n+1)}{2}\right]$

Sum of the elements = 100

53. **(3)**

$$\sum_{r=0}^{25}(4r+1)^{25}C_r = 4\sum_{r=0}^{25} r^{25}C_r + \sum_{r=0}^{25}{}^{25}C_r$$

$$= 4\sum_{r=1}^{25} r \times \frac{25}{r}{}^{24}C_{r-1} + 2^{25} = 100\sum_{r=1}^{25}{}^{24}C_{r-1} + 2^{25}$$

$$= 100.2^{24} + 2^{25} = 2^{25}(50+1) = 51.2^{25}$$

So $k = 51$

54. **(4)**

$3|z_1| = 4|z_2|$

$$\Rightarrow \left|\frac{3z_1}{2z_2}\right| = 2, \text{ Let } \frac{3z_1}{2z_2} = 2\cos\alpha + 2(\sin\alpha)i$$

$$\Rightarrow \frac{8z_2}{3z_1} = 2\cos\alpha - 2(\sin\alpha)i$$

55. **(2)**

One vertex of square is (10(cos α - sin α), 10(sin α + cos α)) and the equation of one of the diagonal is (cos α – sin α)x + (sin α + cos α)y = 10 So the other diagonal can be obtained as (cos α + sin α)x – (cos α – sin α)y = 0 So, the point of intersection of the diagonal will be (5(cos α - sin α), 5(cos α + sin α))

Therefore, the vertex opposite to the given vertex is (0,0).

So. the diagonal length $= 10\sqrt{2}$

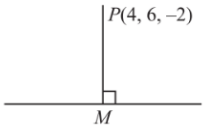
Side length (a) = 10

It is given that

$$a^2 + 11a + 3\left(m_1^2 + m_2^2\right) = 220$$

$$m_1^2 + m_2^2 = \frac{220 - 100 - 110}{3} = \frac{10}{3}$$

56. (4)



Equation of line is $\frac{x+3}{3}=\frac{y-2}{3}=\frac{z-3}{-1}=m$

$M(3m-3,\ 3m+2,\ 3-m)$

D.r's of $PM(3m-7,\ 3m-4,\ 5-m)$

Since, PM is perpendicular to line

$\Rightarrow 3(3m-7)+3(3m-4)-1(5-m)=0$

$\Rightarrow m=2$

$\Rightarrow M(3,8,1) \Rightarrow PM=\sqrt{14}$

57. (4)

$$\text{Mean}=\frac{\sum x_i f_i}{\sum f_i}$$

$$=\frac{0.{}^nC_0+2.{}^nC_1+2^2.{}^nC_2+\ldots+2^n.{}^nC_n}{{}^nC_0+{}^nC_1+\ldots+{}^nC_n}$$

For finding sum of numerator

Let $(1+x)^n={}^nC_0+{}^nC_1x+{}^nC_2x^2+\ldots+{}^nC_nx^n$ ...(i)

Put

$x=2 \Rightarrow 3^n-1=2.{}^nC_1+2^2.{}^nC_2+\ldots+2^n.{}^nC_n$

For sum of denominator

Put $x = 1$ in equation (i)

$2^n={}^nC_0+{}^nC_1+\ldots+{}^nC_n$

$\therefore \frac{3^n-1}{2^n}=\frac{728}{2^n} \Rightarrow 3^n=729 \Rightarrow n=6$

58. (1)

$$\begin{vmatrix} A+6d & 7 & 1 \\ 2(A+1+8d) & 17 & 1 \\ A+2+16d & 17 & 1 \end{vmatrix}+70=0$$

$R_2 \to R_2-R_3$

$\Rightarrow A=-7$

For d, $a = 29$

$A + 6d = 29$

$\Rightarrow d=6,$

So, $a=29, b=42, c=91$

$\therefore c-a-b=20=$ first term

$$S_{20}=\frac{20}{2}\left[2\times 20+19\times\frac{1}{2}\right]$$

59. (4)

We have vertex (5,4) and directrix $3x + y – 29 = 0$ Let foot of perpendicular of (5,4) on directrix be (α, β)

$$\frac{\alpha-5}{3}=\frac{\beta-4}{1}=\frac{-(-10)}{10}$$

$\therefore (\alpha,\beta)=(8,5)$

Since, vertex is midpoint of the foot of perpendicular to the directrix and focus.

So, focus of parabola will be $S = (2,3)$

For any point P(x, y) on the parabola,

$$\Rightarrow (x-2)^2+(y-3)^2=\frac{(3x+y-29)^2}{10}$$

$$\Rightarrow 10\left(x^2+y^2-4x-6y+13\right)=9x^2+y^2+841+6xy-58y-174x$$

$$\Rightarrow x^2+9y^2-6xy+134x-2y-711-0$$

Also equation of the parabola is

$x^2+ay^2+bxy+cx+dy+k=0$

So, $a=9, b=-6, c=134, d=-2, k=-711$

Hence, $a+b+c+d+k=-576$

60. (1)

Given function is

$f(x) = |x –1| \cos (|x – 2|)$

non-diff. at $x = 1$ only

61. (1)

No. is divisible by 3 if sum of its digit is divisible by 3.

(1,1,1) (3,3,3) (5,5,5) (8,8,8)

(5,5,8) (8,8,5) (1,3,5) (1,3,8)

Total number $=1+1+1+1+\frac{3!}{2!}+\frac{3!}{2!}+3!+3!=22$

62. (3)

Given expression,

$\cos^{-1}(\cos(-5))+\sin^{-1}(\sin(6))-\tan^{-1}(\tan(12))$

$=2\pi-5+6-2\pi-(12-4\pi)=4\pi-11$

63. (4)

$$f'(x)=\frac{1}{(x-3)(x-4)}\left(x-\frac{7}{2}\right)>0$$

$$\Rightarrow x\in\left(3,\frac{7}{2}\right)\cup(4,\infty)$$

But $\left(3,\frac{7}{2}\right)$ not in domain

64. (2)

$3 \times 3^9 = 3^{10}$

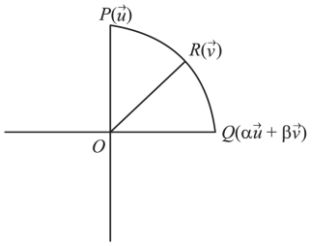
65. (1)

$|\vec{u}|=|\vec{v}|=|\alpha\vec{u}+\beta\vec{v}|$

$(\vec{u})\cdot(\alpha\vec{u}+\beta\vec{v})=0$

$\vec{u}\cdot\vec{v}=|\vec{u}||\vec{v}|\cos 45^\circ$

$$\alpha = -\frac{\beta}{\sqrt{2}}$$



$$= |\alpha\vec{u} + \beta\vec{v}| = r$$

$$\alpha^2 + \beta^2 + \sqrt{2}\alpha\beta = 1$$

$$\alpha = -1, \beta^2 = 2$$

Required equation:

$$x^2 - \left(\alpha + \beta^2\right)x + \alpha\beta^2 = 0$$

$$\Rightarrow x^2 - x - 2 = 0$$

66. (4)

Given $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

We know that $e = \sqrt{1 + \frac{b^2}{a^2}}\, e' = \sqrt{1 + \frac{a^2}{b^2}}$

$$\ell = \frac{2b^2}{a}\ \ell' = \frac{2a^2}{b}$$

$$\left(1 + \frac{b^2}{a^2}\right) = \frac{11}{14} \times \frac{2b^2}{a} \text{ and } \left(1 + \frac{a^2}{b^2}\right) = \frac{11}{8} \times \frac{2a^2}{b}$$

$$7\left(a^2 + b^2\right) = 11 \times b^2\, a \text{ and } 4\left(a^2 + b^2\right) = 11 \times a^2 b$$

$$\frac{7}{4} = \frac{b}{a}$$

67. (1)

$$y(x+1)dx - x^2 dy = 0 \Rightarrow \frac{x+1}{x^2}dx = \frac{dy}{y}$$

$$\left(\frac{1}{x} + \frac{1}{x^2}\right)dx = \frac{dy}{y}$$

Integration both sides, we get

$$\ln|x| - \frac{1}{x} = \ln|y| + c$$

Applying $y(1) = e$, we get

$C = 2$

$$\therefore \ln x - \frac{1}{x} = \ln y - 2$$

$$y = e^{\left(\ln x - \frac{1}{x} + 2\right)}$$

$$\therefore \lim_{x \to 0^+} e^{\left(\ln x - 1\frac{1}{x} + 2\right)}$$

$$= e^{-\infty} = 0$$

68. (2)

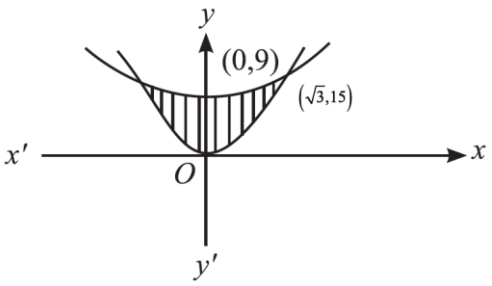
Required Area of shaded region from the fig.

$$= 2\int_0^{\sqrt{3}} \left(2x^2 + 9 - 5x^2\right)dx$$

$$= 2\int_0^{\sqrt{3}} \left(9 - 3x^2\right)dx$$

$$= \int_0^{\sqrt{3}} \left(3 - x^2\right)dx = 6\left[3x - \frac{x^3}{3}\right]_0^{\sqrt{3}}$$

$$= 6\left[\frac{9\sqrt{3} - 3\sqrt{3}}{3}\right] = 12\sqrt{3}$$

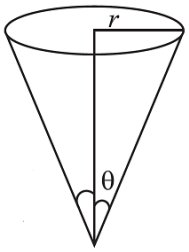


69. (2)

Let radius of cone is r and height is h

$$\tan\theta = \frac{1}{2} = \frac{r}{h}$$

$$\Rightarrow r = \frac{h}{2}$$



Now, volume $(V) = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\frac{h^3}{4}$ $\tan\alpha = \frac{1}{2}$

$$\frac{dV}{dt} = \frac{\pi}{12}(3h)^2\left(\frac{dh}{dt}\right)$$

$$5 = \frac{\pi}{4}(100)\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{5\pi}$$

70. (3)

Given Binomial $\left(x - \frac{3}{x^2}\right)^n, x \neq 0, n \in N,$

Sum of coefficients of first three terms

$${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 3^2 = 376$$

$$\Rightarrow 3n^2 - 5n - 250 = 0$$

$$\Rightarrow (n - 10)(3n + 25) = 0$$

$$\Rightarrow n = 10$$

Now general term $T_{r+1} = {}^{10}C_r x^{10-r}\left(\frac{-3}{x^2}\right)^r$

$= {}^{10}C_r x^{10-r}(-3)^r \cdot x^{-2r}$

$= {}^{10}C_r (-3)^r \cdot x^{10-3r}$

Coefficient of $x^4 \Rightarrow 10-3r=4$

$\Rightarrow r=2$

Hence, coefficient of x^4

$= {}^{10}C_2(-3)^2 = 405$

71. (63)

$$\int\left(x^{20}+x^{13}+x^6\right)\left(2x^{21}+3x^{14}+6x^7\right)^{1/7}dx$$

$2x^{21}+3x^{14}+6x^7 = t$

$42\left(x^{20}+x^{13}+x^6\right)dx = dt$

$$\frac{1}{42}\int_0^{11} t^{\frac{1}{7}}dt = \left(\frac{t^{\frac{8}{7}}}{\frac{8}{7}}\times\frac{1}{42}\right)_0^{11}$$

$$= \frac{1}{48}\left(t^{\frac{8}{7}}\right)_0^{11} = \frac{1}{48}(11)^{8/7}$$

$l=48, m=8, n=7$

$l+m+n=63$

72. (10)

Given $x^2+y^2-2\sqrt{2}x-6\sqrt{2}y+14=0$

Centre $=(\sqrt{2},3\sqrt{2})$

Radius $=\left((\sqrt{2})^2+(3\sqrt{2})^2-14\right)^{1/2}$

$=(2+18-14)^{1/2}=\sqrt{6}$

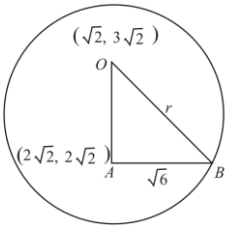
Now, $(x-2\sqrt{2})^2+(y-2\sqrt{2})^2=r^2$

Centre $=(2\sqrt{2},2\sqrt{2})$

$OA=\sqrt{(2\sqrt{2}-\sqrt{2})^2+(2\sqrt{2}-3\sqrt{2})^2}$

$=\sqrt{2+2}=2$

$r^2=(\sqrt{6})^2+(2)^2=6+4=10$



73. (40)

$2^m - 2^n = 112 \Rightarrow m = 7$ & $n = 4$

74. (10)

Total ways = 6!

Ways satisfying $g(3) = 2g(1)$ is 3

Number of onto function $3 \times 4!$

Hence, probability = $\frac{1}{10}$

75. (1)

Given trigonometric equation is

$$\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$$

$$=\left(\cos\left(\frac{2\pi}{7}\right)+\cos\left(\frac{6\pi}{7}\right)\right)+\cos\left(\frac{4\pi}{7}\right)$$

$$=\cos\left(\frac{4\pi}{7}\right)\left[2\cos\left(\frac{2\pi}{7}\right)+1\right]$$

$$=\cos\left(\frac{4\pi}{7}\right)\left[2\left(1-2\sin^2\left(\frac{\pi}{7}\right)\right)+1\right]$$

$$=\cos\left(\frac{4\pi}{7}\right)\left[3-4\sin^2\left(\frac{\pi}{7}\right)\right]=\frac{\sin\left(3\times\frac{\pi}{7}\right)}{\sin\frac{\pi}{7}}\times\cos\left(\frac{4\pi}{7}\right)$$

$$=\frac{2\sin\left(\frac{3\pi}{7}\right)}{2\sin\frac{\pi}{7}}\times\cos\left(\frac{4\pi}{7}\right)$$

Apply $2\sin A\cos B = \sin(A+B)+\sin(A-B)$

$$=\frac{\sin\left(\frac{7\pi}{7}\right)+\sin\left(\frac{-\pi}{7}\right)}{2\sin\frac{\pi}{7}},\ \text{Here, } \sin(\pi)=0.$$

$$=\frac{-\sin\frac{\pi}{7}}{2\sin\frac{\pi}{7}}=-\frac{1}{2}$$



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